

Iterated Symmetric Centipede:
A Five-Strategy Evolutionary Analysis
All- M , All-1, Grim, Grim-Patient, Grim-Forgiver

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The One-Shot Centipede Game

Setup

- Two players alternate along a path of M nodes.
- At each node the moving player can **Pass** or **Take**.
- Taking ends the game; total payoff at node t is t .
- The taker receives share $p \in (\frac{1}{2}, 1]$; the other gets $1 - p$.

The paradox

Backward induction $\Rightarrow$ take at node 1 (SPE).

Yet (ρ_1, ρ_1) is Pareto-dominated by (ρ_M, ρ_M) .

Symmetric version (OSC)

Randomise who moves first $\Rightarrow$ symmetric expected payoffs. Strategy ρ_m : “take at the first opportunity at or after node m ”.

OSC payoff matrix ($M=4$, $p=3/4$):

$$A = \begin{pmatrix} 1 & 9/4 & 9/4 & 9/4 \\ 3/4 & 2 & 15/4 & 15/4 \\ 3/4 & 5/4 & 3 & 21/4 \\ 3/4 & 5/4 & 7/4 & 4 \end{pmatrix}$$

Five Strategies in the Iterated Game (ISC)

The ISC repeats the OSC for T rounds; players observe full history.

All- M Always play ρ_M . Full cooperator; never punishes.

All-1 Always play ρ_1 . Full defector; never cooperates.

Grim Start with ρ_M . At the *first* defection by the opponent, switch to ρ_1 *permanently*.

Grim-Patient Like Grim, but requires *two consecutive* defections before triggering.

Grim-Forgiver Punishes with ρ_1 for exactly *two rounds* after each defection, then unconditionally returns to ρ_M — repeating the loop indefinitely.

Key observation: All- M , Grim, GrimP, GrimF never trigger each other — they all play ρ_M throughout whenever paired together.

The ISC Payoff Matrix ($M=4$, $T=10$, $p=3/4$)

$$C = \begin{pmatrix} 40 & 7.5 & 40 & 40 & 40 \\ 22.5 & 10 & 11.25 & 12.5 & 15 \\ 40 & 9.75 & 40 & 40 & 40 \\ 40 & 9.5 & 40 & 40 & 40 \\ 40 & 9.0 & 40 & 40 & 40 \end{pmatrix}$$

Rows/columns: All- M , All-1, Grim, GrimP, GrimF.

Three structural facts:

- 1 All cooperating strategies earn $TM = 40$ against *each other*. They are **payoff-equivalent** when All-1 is absent.
- 2 All-1's payoff against a cooperator depends on how long that cooperator holds out before switching.
- 3 GrimF is **most exploitable**: it cycles back to ρ_M every 3 rounds, so All-1 earns $C(2, 5) = 15 > C(2, 4) = 12.5 > C(2, 3) = 11.25$.

Nash Equilibria

Cooperative equilibrium set

Any mixture over $\{\text{All-M}, \text{Grim}, \text{GrimP}, \text{GrimF}\}$ is a Nash equilibrium.

This is a *continuum* of NE — a direct consequence of the payoff degeneracy $C(i,j) = TM$ for all cooperating pairs.

All-1 equilibrium

(σ_2, σ_2) is a NE if and only if $p \geq 2/3$.

The critical threshold $p^* = 2/3$

- $p < 2/3$: defecting is not profitable; cooperation dominates.
- $p > 2/3$: the first-mover advantage is large enough for All-1 to thrive.

Evolutionarily Stable Strategies (ESS)

- **No pure cooperator is an ESS.**

Because all cooperators earn identical payoffs against each other, none can resist a neutral mutation into another cooperating strategy.

- **No mixed strategy on the cooperative face is an ESS.**

Same reason: the cooperative face $\mathcal{F} = \{x_2 = 0\}$ is neutrally stable, not strictly stable.

- **All-1 is an ESS if and only if $p > 2/3$.**

Below the threshold, any cooperating strategy can invade an all-defector population and grow.

Implication: below p^* , the system converges to cooperation but the final mix among cooperators is determined by initial conditions and drift — not by selection.

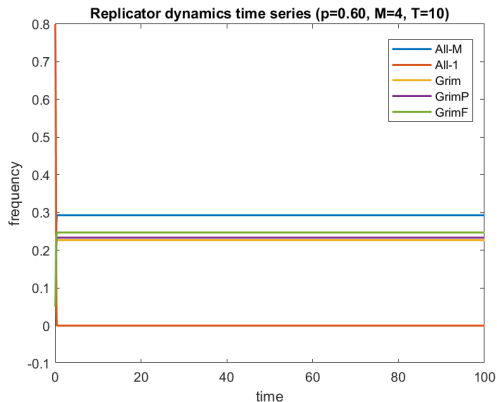
Large-population, continuous-time model. Strategy m grows if its payoff exceeds the population average:

$$\dot{x}_m = x_m [(Cx)_m - \mathbf{x}^\top C\mathbf{x}]$$

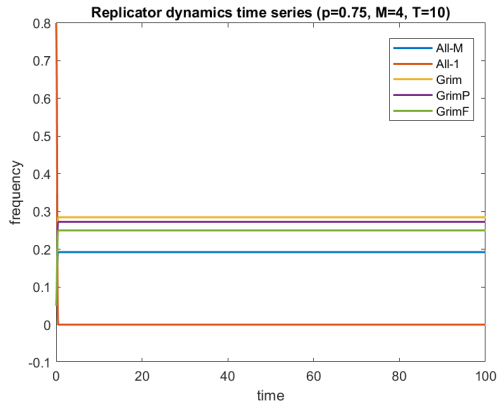
Key results:

- The entire cooperative face $\mathcal{F} = \{x_2 = 0\}$ is a *rest-point set* — no drift among cooperators.
- The only interesting dynamics are in the x_2 (All-1) direction.
- $\mathcal{F}$ is asymptotically stable (w.r.t. All-1 invasion) iff $p < 2/3$.

Replicator Dynamics — Time Series



$p = 0.60 < p^*$: All-1 driven to zero.

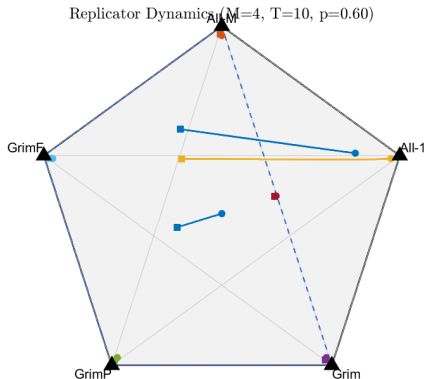


$p = 0.75 > p^*$: All-1 remains; cooperators squeezed.

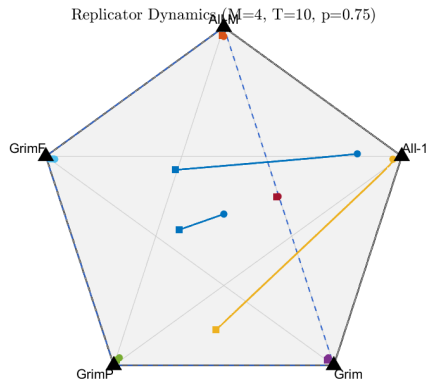
Note: equilibrium frequencies are essentially flat — the dynamics settle very quickly.

Replicator Dynamics — Phase Portraits

Pentagon projection: each vertex = one pure strategy; interior = mixed population.

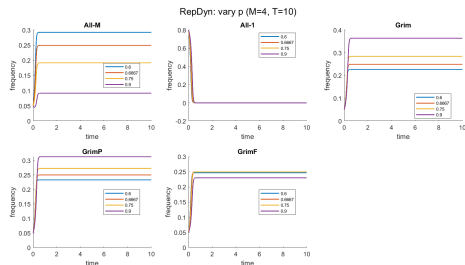


$p = 0.60$: orbits converge to the cooperative face (away from All-1 vertex).

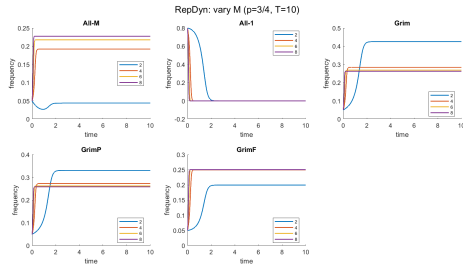


$p = 0.75$: orbits converge toward the All-1 vertex.

Replicator Dynamics — Varying Parameters

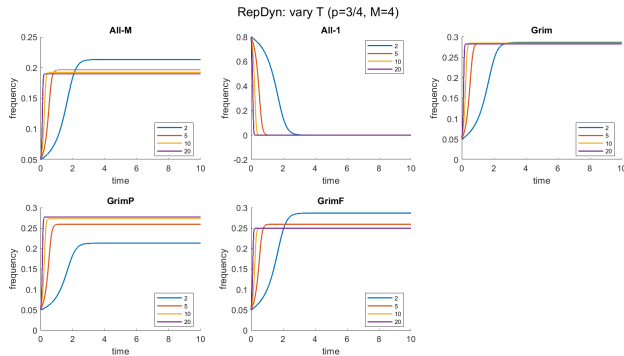


Vary p : threshold at $p^* = 2/3$ clearly visible.



Vary M : larger centipede amplifies cooperation incentives.

Replicator Dynamics — Effect of T



More repetitions (T) mainly affects **convergence speed**, not the equilibrium frequencies. The long-run outcome is determined by p alone.

Markov Chain — Finite Population Model

Setup: N agents, each carrying one pure strategy.

At each step: one agent compares payoff with a random opponent and imitates with probability proportional to the payoff difference (*Pairwise Proportional Imitation*).

State space: $(s_1, s_2, s_3, s_4, s_5)$ with $\sum s_i = N$.

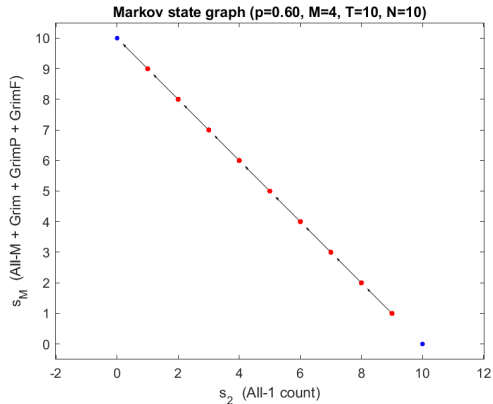
Reduced to 2D: $s_M = s_1 + s_3 + s_4 + s_5$ (cooperator count) vs. s_2 (All-1 count).

Absorbing states

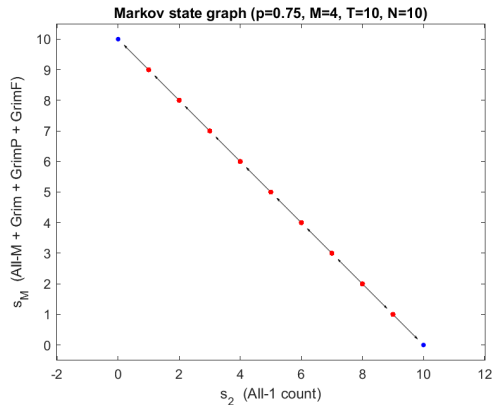
- The entire **cooperative face** $\mathcal{A} = \{s_2 = 0\}$: any combination of cooperators with no All-1 agents is absorbing.
- The **pure All-1** state $(0, N, 0, 0, 0)$.

All other states are transient.

Markov Chain — State Graphs

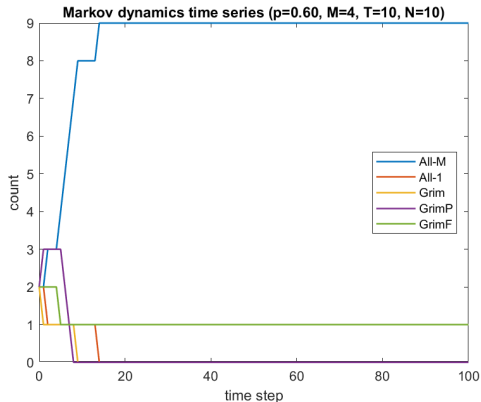


$p = 0.60$: drift pushes system toward cooperative face (blue dots = absorbing, red = transient).

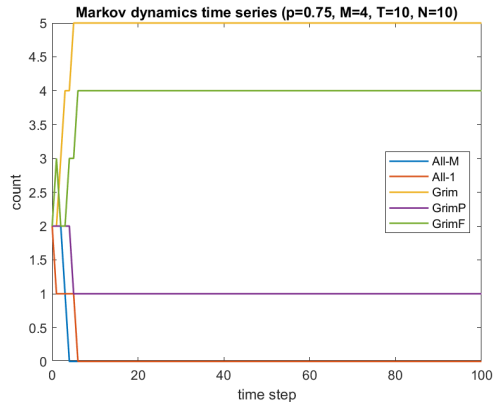


$p = 0.75$: both cooperative face and All-1 vertex are attractors.

Markov Chain — Time Series

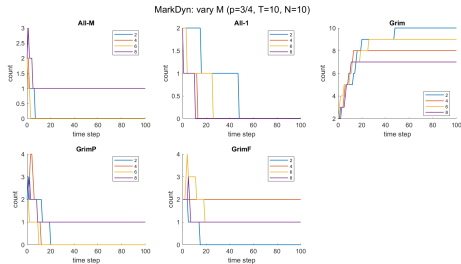


$p = 0.60$: All-M dominates; others eliminated.

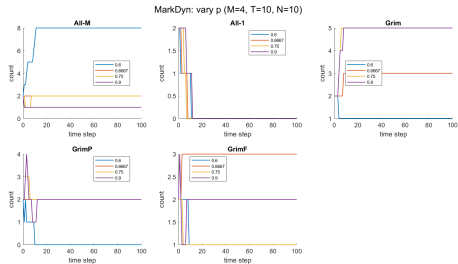


$p = 0.75$: GrimF and Grim persist; All-M and All-1 collapse.

Markov Chain — Varying Parameters



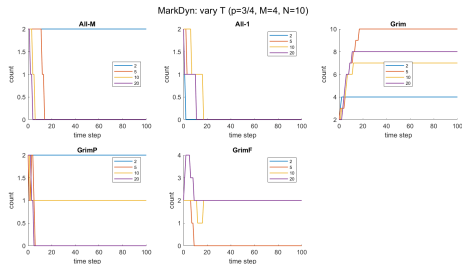
Vary M



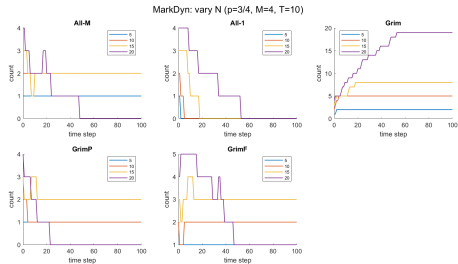
Vary p

Larger M strengthens cooperation. Threshold $p^* = 2/3$ again separates the two regimes.

Markov Chain — Varying T and N



Vary T : more rounds stabilise cooperation.



Vary N : larger population slows drift but same long-run outcome.

Comparing the Three Grim Variants

All three earn TM against cooperators — they differ only in how exploitable they are by All-1.

Strategy	Trigger condition	All-1 earns	
Grim	1st defection (permanent)	$C(2, 3) = 11.25$	least exploitable
Grim-Patient	2 consecutive (permanent)	$C(2, 4) = 12.5$	
Grim-Forgiver	1st defection, forgive after 2	$C(2, 5) = 15.0$	most exploitable

Why GrimF is most exploitable:

GrimF cycles $[\rho_M, \rho_1, \rho_1, \rho_M, \rho_1, \rho_1, \dots]$ when facing All-1. It returns to full cooperation every 3 rounds, allowing All-1 to exploit it repeatedly.

Trade-off: Grim is harshest and least exploitable, but GrimF allows partial cooperation recovery after punishment — a softer but more vulnerable approach.

- 1 **Payoff degeneracy.** All cooperating strategies earn TM against each other $\Rightarrow$ continuum of Nash equilibria among cooperators.
- 2 **Critical threshold** $p^* = 2/3$. Cooperation survives evolution if and only if $p < 2/3$.
- 3 **No ESS among cooperators.** Neutral drift, not selection, determines the long-run mix of All- M / Grim / GrimP / GrimF.
- 4 **GrimF is most exploitable.** Its periodic forgiveness allows All-1 to extract full cooperation repeatedly. Grim is the most resistant to exploitation.
- 5 **Markov dynamics.** The absorbing face $\mathcal{A} = \{s_2 = 0\}$ captures the long-run cooperative outcome. Convergence to cooperation or All-1 depends on p and initial conditions.
- 6 **Learning interpretation.** PPI $\equiv$ social imitation; replicator $\equiv$ reinforcement learning. Both frameworks converge to the same p^* -threshold behavior.

- Add **Tit-for-Tat** and **Generous-Tit-for-Tat** to the strategy set.
- Study the effect of **noise/mutation** on the neutrally stable cooperative face.
- Analyse **asymmetric populations** via two-population replicator dynamics.

Thank you — Questions?